

MICROBIOME

Comprehensive analysis of microbial content in whole-genome sequencing samples from The Cancer Genome Atlas project

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In recent years, a growing number of publications have reported the presence of microbial species in human tumors and mixtures of microbes that appear to be highly specific to different cancer types. Our recent reanalysis of data from three cancer types revealed that technical errors have caused erroneous reports of numerous microbial species found in sequencing data from The Cancer Genome Atlas (TCGA) project. Here, we have expanded our analysis to cover all 5734 whole-genome sequencing (WGS) datasets currently available from TCGA, covering 25 distinct types of cancer. We analyzed the microbial content using updated computational methods and databases and compared our results to those from two major recent studies that focused on bacteria, viruses, and fungi in cancer. Our results expand upon and reinforce our recent findings, which show that the presence of microbes is far smaller than had been previously reported and that many species identified in TCGA data might not be present at all. As part of this expanded analysis and to help others avoid being misled by flawed data, we have released a dataset that contains detailed read counts for bacteria, viruses, archaea, and fungi detected in all 5734 TCGA samples, which can serve as a public reference for future investigations.

INTRODUCTION

A number of recent studies have used the vast sequencing resource created by The Cancer Genome Atlas (TCGA) project to explore the potential role of microbial species in cancer. Although most TCGA data were collected with the goal of studying human genetic variation or gene expression, microbes present in the tumors, including viruses, bacteria, and fungi, might also be captured as an incidental side effect of sequencing experiments. Identifying microbes in a human tumor sample, in which the vast majority of the biomass is expected to be human in origin, requires great care to not be misled by contaminants, sequencing vectors, or other artifacts that might also be present in the data. In this study, our objective was to conduct an exhaustive and meticulous survey of microbial communities across thousands of whole-genome sequencing (WGS) samples from the TCGA project, with the goal of identifying microbes within these samples. By making our results publicly available, we hope to spur additional research that may amplify or alternatively refute recent findings of microbiomes in a wide variety of tumor types.

We also compared our findings with two recent studies that used much of the same TCGA data and described findings that were, in some instances, substantially affected by contamination. Those studies and others that have relied on their data have implicated the microbiome in various aspects of cancer, from modulating the tumor microenvironment to influencing treatment responses. In the first study, Poore *et al.* (1) analyzed 17,625 samples from TCGA and reported that they were able to use machine learning algorithms to construct highly discriminative microbial signatures in 32 types of cancer. Their classifiers, which used combinations of bacteria,

archaea, and viruses, were remarkably accurate, obtaining 95 to 100% accuracy at discriminating each of the 32 cancer types against all others. They reported additional models with similar accuracy at distinguishing tumors from matched normal samples in 15 cancer types [because of our findings (2), the journal *Nature* retracted the Poore *et al.* study in July of 2024 (3)]. In the second study, Narunsky-Haziza *et al.* (4) analyzed 17,401 samples from the TCGA project and other sources with the goal of identifying associations between cancer and fungi, which the first study had not considered. They reported distinctive fungal signatures of cancer in most of the 35 cancer types they considered (4). Our analysis here looked at many of the same TCGA samples in an effort to replicate some of these findings.

One source of overcounts when analyzing human samples for microbial content is data contamination in public genome databases. As described previously, the inadvertent inclusion of human DNA within microbial genomes has affected thousands of genomes (5). When creating a microbial sequence database, it is crucial to take rigorous steps to remove these computational contaminants, which otherwise may substantially skew the results of metagenomic analyses. Such contamination events can be especially problematic when working with low-biomass samples, where the microbial content is expected to be a very small proportion of the total sample. This is precisely the scenario encountered when “mining” human DNA sequencing projects such as TCGA for microbial content.

We should emphasize further that even a tiny amount of contamination in a genome’s database sequence can lead to enormous overcounts of bacterial species for the following reason: The most common source of human contamination in bacterial genome databases is high-copy human repeats such as Alus, long interspersed elements, and short interspersed elements, as described earlier (5). Any WGS sample from human tissue is likely to contain large numbers of reads from these widespread repeats. Thus, if one scans a human DNA sample against a bacterial genome that is contaminated with even one of these human repeats, many human reads will appear to

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match the bacterium. For a typical TCGA sample, it would not be unexpected to find tens or even hundreds of thousands of reads in-correctly matching a bacterial genome in this circumstance (2).

Vector contamination, in which reads deriving from vectors such as manufacturer-specific sequencing primers make their way into a genome assembly, further compounds the challenges associated with metagenomic analyses. Sequences originating from vectors have inadvertently found their way into genome databases, where they might be labeled as bacteria, fungi, plants, or animals. As we describe below, some fungal genome sequences in public databases are contaminated with vector or adaptor sequences, which can lead to large numbers of false-positive matches to a sample that was sequenced using the same vectors.

RESULTS

We analyzed the microbial content of 5734 WGS samples from TCGA, which comprised all of the available WGS samples as of late

2023. Although TCGA also includes large numbers of RNA sequencing (RNA-seq) experiments, we excluded them because they used poly(A) selection to capture messenger RNA. Bacterial transcripts do not have long poly(A) tails (6) and will not be captured, except rarely, with poly-A selection protocols. Thus, any bacterial sequences found in a human RNA-seq experiment are likely to represent a small, highly biased sample of transcripts that happen to contain a poly-A subsequence, and the inclusion of human RNA-seq data in a search for bacterial signatures, as has been done in previous studies (1), is not advisable.

We removed human sequences from the TCGA data by mapping the reads against both the GRCh38 reference genome and the CHM13 human genome. As shown in Table 1, after removal of human sequences, the number of reads remaining in most primary tumor samples was relatively small, averaging 2.4 million reads per sample (0.35% of the total). Across all primary tumor samples, a total of 6.5 billion reads remained after two-pass filtering. Of these, we identified 323 million as human and 986 million as vector contaminants during

Table 1. Overview of 2708 primary tumor samples of the 5734 WGS samples analyzed in this study. Column 4 shows how many reads remained per sample after removing reads that mapped to the GRCh38 human genome. Column 6 shows the number of reads remaining after a second pass further removed reads matching the CHM13 genome. Columns 8 and 10 show the average (avg) numbers of reads from column 6 that were identified by Kraken as either human or vector.

Cancer type	Total no. samples	Average read count (millions)	Unmapped reads after mapping to GRCh38 (avg, thousands)		Unmapped reads after mapping to GRCh38 + CHM13 (avg, thousands)		Kraken-identified human reads (avg, thousands)		Kraken-identified vector reads (avg, thousands)	
BLCA	140	304	4474	(1.47%)	3498	(1.15%)	164.1	(0.05%)	15	(0.00%)
BRCA	120	826	2575	(0.31%)	2276	(0.28%)	38.0	(0.00%)	555	(0.07%)
CESC	64	433	2754	(0.64%)	2186	(0.50%)	105.6	(0.02%)	98	(0.02%)
COAD	131	417	5803	(1.39%)	5089	(1.22%)	211.1	(0.05%)	792	(0.19%)
DLBC	7	1213	478	(0.04%)	478	(0.04%)	0.2	(0.00%)	454	(0.04%)
ESCA	57	392	3224	(0.82%)	2389	(0.61%)	83.1	(0.02%)	522	(0.13%)
GBM	54	795	1559	(0.20%)	1302	(0.16%)	224.7	(0.03%)	21	(0.00%)
HNSC	171	434	4205	(0.97%)	3413	(0.79%)	181.7	(0.04%)	295	(0.07%)
KICH	50	1124	582	(0.05%)	582	(0.05%)	0.3	(0.00%)	556	(0.05%)
KIRC	44	824	1257	(0.15%)	1257	(0.15%)	0.1	(0.00%)	453	(0.05%)
KIRP	38	1109	442	(0.04%)	442	(0.04%)	0.3	(0.00%)	422	(0.04%)
LAML	64	721	1986	(0.28%)	1575	(0.22%)	57.4	(0.01%)	814	(0.11%)
LGG	83	468	4610	(0.99%)	3670	(0.78%)	92.1	(0.02%)	2	(0.00%)
LIHC	54	1213	531	(0.04%)	530	(0.04%)	0.5	(0.00%)	497	(0.04%)
LUAD	289	565	4569	(0.81%)	3813	(0.67%)	147.8	(0.03%)	1453	(0.26%)
LUSC	50	1065	200	(0.02%)	199	(0.02%)	0.1	(0.00%)	2	(0.00%)
OV	58	935	277	(0.03%)	276	(0.03%)	0.4	(0.00%)	194	(0.02%)
PRAD	136	328	6364	(1.94%)	4891	(1.49%)	204.1	(0.06%)	9	(0.00%)
READ	60	325	7186	(2.21%)	6218	(1.91%)	247.1	(0.08%)	1041	(0.32%)
SARC	40	906	217	(0.02%)	217	(0.02%)	0.2	(0.00%)	190	(0.02%)
SKCM	19	311	2369	(0.76%)	1800	(0.58%)	102.1	(0.03%)	21	(0.01%)
STAD	149	415	4668	(1.13%)	3302	(0.80%)	226.2	(0.05%)	18	(0.00%)
THCA	619	1087	568	(0.05%)	435	(0.04%)	18.2	(0.00%)	22	(0.00%)
UCEC	160	434	4333	(1.00%)	3537	(0.81%)	405.0	(0.09%)	375	(0.09%)
UVM	51	145	2820	(1.95%)	1905	(1.32%)	83.8	(0.06%)	2	(0.00%)
Total	2708	1,845,742	8,082,762	(0.44%)	6,531,256	(0.35%)	322,540	(0.02%)	985,611	(0.05%)
Average	108	682	2985	(0.44%)	2412	(0.35%)	119	(0.02%)	364	(0.05%)

classification with Kraken. Data files S1 and S2 list the same statistics for 569 solid-tissue normal and 2277 blood-derived normal samples, in which the average proportion of microbial reads after removal of human sequences was 0.57 and 0.73%, respectively.

We used KrakenUniq (7) to classify the filtered reads against a microbial database containing 50,651 genomes representing 30,355 species including bacteria, archaea, viruses, and fungi. Across all samples, we identified 11,388 noneukaryotic microbial species that occurred at least once. Data file S3 reports the number of sequencing reads detected for every species in each of the samples. Data file S4 includes normalized counts for the data in data file S3, where read counts are divided by the total number of reads sequenced for each sample, in millions. The vast majority of the entries in these tables are zero: Of ~65 million entries, 1,889,454 (2.9%) have non-zero values, and 215,302 (0.3%) have raw read counts of 10 or greater. We then classified reads that were not identified in the previous step against a comprehensive fungal genome database containing 557 species. Data file S5 reports the raw read counts for all 5734 samples for each of these fungal species, with normalized counts reported in data file S6.

Even after alignment against two human genomes, an average of ~119,000 reads per sample were still classified as human by the Kraken program (data file S1). This highlights the necessity of including the human genome in any metagenomics database, even if the data were prefiltered to remove human reads. Figure 1 shows the breakdown of bacterial, viral, and fungal read counts for 2708 primary tumor samples across the 25 cancer types.

Of note, some samples contained, among the various microbes found, individual species with unusually high read counts. For example, although only a handful of samples contained more than 100 reads from *Helicobacter pylori*, one sample contained >175,000 reads. That sample was from stomach cancer [stomach adenocarcinoma (STAD)] data, but the source was normal tissue rather than a tumor. For *Fusobacterium nucleatum*, which has been associated with colon cancer, only 123 samples had >100 reads, and three samples had >100,000 reads, all found in tumor samples from head and neck cancer [head and neck squamous cell carcinoma (HNSC)]. A total of 105 of the 123 samples were from tumors, and 87 of those were either oral cavity or colon tissues. Although a closer analysis of these and other observations is beyond the scope of this study, the data provided in our supplementary tables should provide a solid basis for further investigations.

The TCGA project used bwa (8) to align all reads to the human genome and also to a small number of viruses, including multiple subtypes of human papillomavirus (HPV), hepatitis B and C, and cytomegalovirus. Because these reads counts were based on alignments from bwa and not KrakenUniq and also because TCGA separated out the HPV matches into their subtypes, we provide those counts separately in data file S7. Data files S3 and S4 include species-level read counts for all of these viruses as well.

A previous study of TCGA data reported that the cervical cancer [cervical squamous cell carcinoma (CESC)] dataset had a high proportion of samples positive for HPV, in particular HPV-16 and HPV-18 (9). Although the earlier study relied on polymerase chain reaction (PCR) for detection, which should be more sensitive than WGS, our analysis of the WGS data confirmed those findings. Among the 66 nontumor CESC samples, nearly all had very low counts for all subtypes of HPV, with only one sample containing >1000 reads for HPV-16. In contrast, among the 64 CESC tumor

samples, 31 samples had >1000 reads from HPV-16, 8 other samples had >1000 reads from HPV-18, and 11 more had >1000 reads from other HPV strains (data file S7). Also worth noting is that 33 of 171 tumor samples in HNSC had >1000 reads from HPV-16, whereas all of the normal samples from HNSC had fewer than 200 HPV-16 reads. HPV-16 is formally classified under the species *Alphapapillomavirus 9*, which as shown in data file S3 was found in 131 CESC and HNSC samples.

Abundant bacteria, viruses, and fungi may be contaminants

The most abundant bacterial and viral species across all 5734 samples, in our analysis, were *Delftia acidovorans*, *Lymphocryptovirus humangamma4* (also known as Human herpesvirus 4), *Rothia mucilaginosa*, *Pseudomonas* sp. J380, *Human mastadenovirus C*, *Escherichia coli*, *Bacteroides fragilis*, *Prevotella intermedia*, and *Alphapapillomavirus 9* (HPV). Each of these is known to be present on human skin, in the oral cavity, or in the gut, and each was found in large numbers of both tumor and normal samples, ranging from 378 samples containing *Human mastadenovirus C* to 1968 samples with *D. acidovorans* and 4440 samples with *E. coli*. Although species such as *B. fragilis* might truly derive from human tumors, a possible explanation for the presence of some species, especially for tissue samples from internal organs not known to harbor a microbiome, is that they might represent contamination, with possible sources including other tissues within the human body or the sequencing process itself. For example, the most frequently observed fungal species, appearing in 2312 samples, was *Saccharomyces cerevisiae*, a widely used model organism that is not a human pathogen and that frequently appears as a cross-contaminant in sequencing centers. Another example is the bacterium *Cutibacterium acnes*, a common human skin bacterium that was abundant in most of the cancer types. Another abundant species was *Rosellinia necatrix partitivirus 8*, a virus that infects plant fungi and has no known associations with human disease (10, 11).

Associations between microbes and cancer types

To determine whether any of the microbial species detected in our analysis might be associated with cancer, we examined the most abundant microbes in each of the cancer types, using the normalized values in data file S4, and also looked specifically for bacteria and viruses that have known associations with cancer. The top species for each cancer type are shown in fig. S1.

After normalization, a few known cancer-associated microbes appeared relatively abundant; for example as mentioned above, HPV was identified in most of the cervical cancer samples and in some of the head and neck cancers. We also found that *B. fragilis* was the top species for rectal adenocarcinoma (READ) and the fourth-most abundant microbe in colon adenocarcinoma (COAD), and *H. pylori* was the 17th-most common microbe in STAD.

Comparison of bacterial and viral read counts to previous reports

In a recent reanalysis (2) of TCGA sequence data from three cancer types, bladder cancer, breast cancer, and head and neck cancer, we reported that an earlier study [now retracted (3)] by Poore *et al.* of the same data (1) had described read counts that were much higher. In particular, we found that >95% of the read counts in the earlier study were inflated by at least a factor of 10 (2). Using the more comprehensive data reported here, we can now confirm the previous findings and extend them to all 25 cancer types for which WGS data are available.

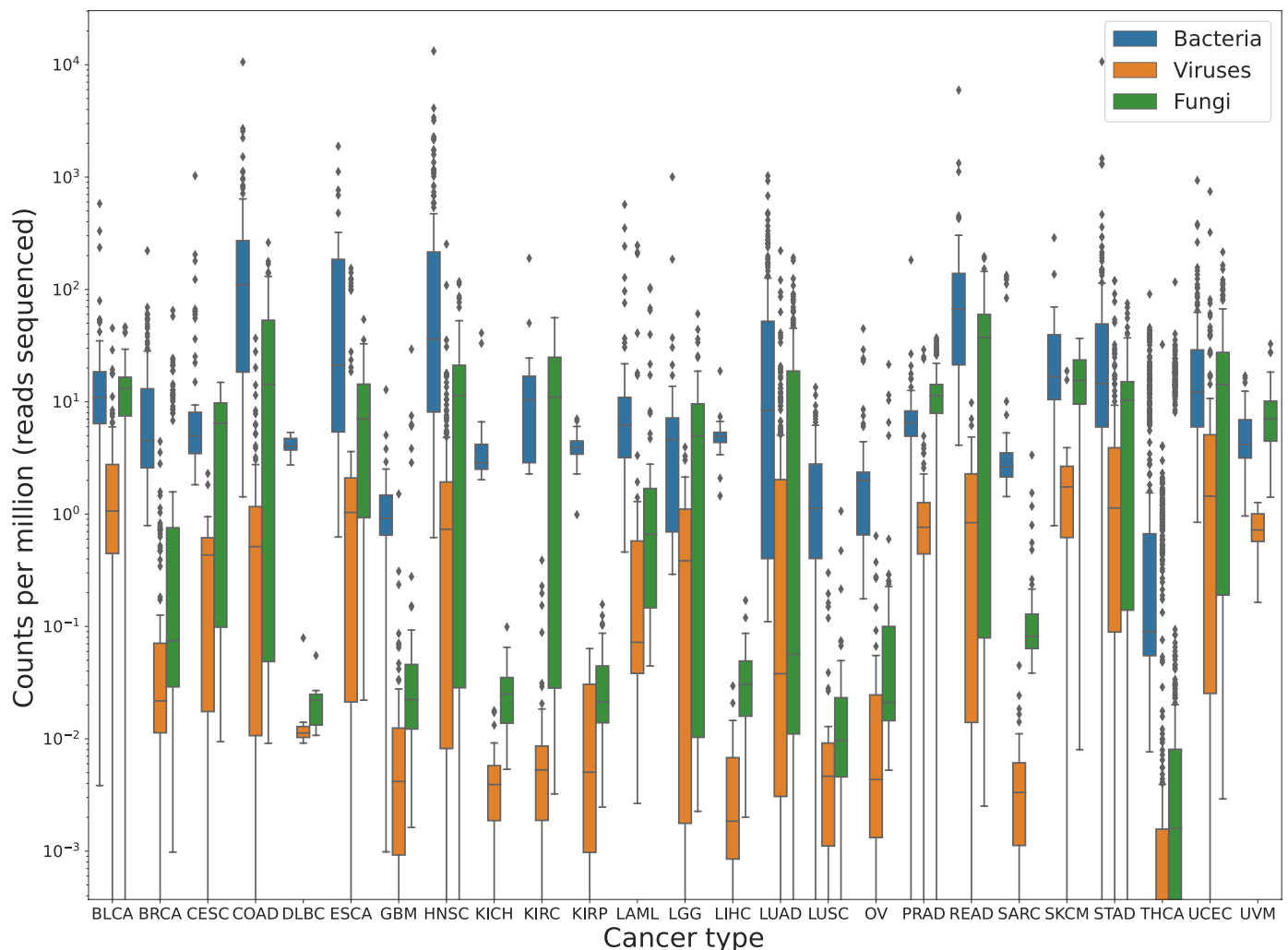


Fig. 1. Microbial content found in 2708 primary tumor samples across 25 cancer types. Box plots show the interquartile range of read counts for the samples from each cancer type, with a horizontal line indicating the mean value. On average, bacteria (blue) had much higher read counts than fungi (green) or viruses (orange). Read counts were normalized by the number of reads sequenced in each sample, in millions; in other words, a value of 10 indicates 10 reads for every million reads in the sequencing run. No adjustments were made for genome size here, which explains the lower read counts for viruses. For clarity, the y axis uses a log scale. LAML, acute myeloid leukemia; ACC, adrenocortical carcinoma; BLCA, bladder urothelial carcinoma; LGG, brain lower-grade glioma; BRCA, breast-invasive carcinoma; CHOL, cholangiocarcinoma; LCML, chronic myelogenous leukemia; CNTL, controls; ESCA, esophageal carcinoma, FPPP, FFPE Pilot Phase II; GBM, glioblastoma multiforme; KICH, kidney chromophobe; KIRC, kidney renal clear cell carcinoma; KIRP, kidney renal papillary cell carcinoma; LIHC, liver hepatocellular carcinoma; DLBC, lymphoid neoplasm diffuse large B cell lymphoma; MESO, mesothelioma; MISC, miscellaneous; OV, ovarian serous cystadenocarcinoma; PAAD, pancreatic adenocarcinoma; PCPG, pheochromocytoma and paraganglioma; PRAD, prostate adenocarcinoma; SARC, sarcoma; SKCM, skin cutaneous melanoma; TGCT, testicular germ cell tumors; THYM, thymoma; THCA, thyroid carcinoma; UCS, uterine carcinosarcoma; UCEC, uterine corpus endometrial carcinoma; UVM, uveal melanoma.

The Poore *et al.* study (1) reported read counts for 1993 microbial genera, including bacteria, archaea, and viruses. Of the 5734 WGS samples analyzed in this study, we identified 4550 samples that matched those from the Poore *et al.* study; the remaining samples were added to TCGA more recently and thus were excluded from this comparison. We compared genera in which either study identified at least 10 reads in at least one of the 4550 samples. Our analysis found 1507 bacterial genera satisfying this condition, whereas Poore *et al.* found 1524, of which 876 genera were found in both studies (Fig. 2A). We also identified 32 archaeal and 104 viral genera, compared with 95 and 204 reported by Poore *et al.* (Fig. 2, B and C).

Data files S8 and S9 contain read counts for all 4550 samples and for the 3563 genera found in either study, with data file S8 containing the results for this study and data file S9 showing the corresponding counts from Poore *et al.* (1). We compared the findings by computing the ratios of read counts (S9/S8), replacing any zero values with 1 to avoid division by zero. We then examined the ratios of read counts for all cells with a count ≥ 10 in at least one study, which comprised 2,197,730 entries.

This analysis showed that the microbial read counts reported by Poore *et al.* were vastly higher than the counts found here. The median ratio was 56; that is, half of the read counts in the earlier study were at least 56 times higher, and 90% of the values were at least 11

times higher. The top 5% (109,887) of the read count ratios were more than 9186-fold higher (Fig. 3). The primary reason for these overestimates, as explained previously (2), was the use of a database containing thousands of incomplete (“draft”) bacterial genomes, which themselves were contaminated with human sequences, resulting in millions of human reads in TCGA being mistaken for bacterial reads.

To provide another comparison, we identified the top 10 most-abundant microbial genera across all 4550 WGS samples reported in (1) and compared them with our read counts for the same genera.

Fig. 2. Comparison of bacterial, viral, and archaeal genera. Comparisons were made between genera found in 4550 TCGA samples versus those found in an earlier study by Poore *et al.* (1). A genus was counted only if at least one sample had at least 10 reads. (A) Counts of bacterial genera; (B) counts of archaeal genera; (C) counts of viral genera.

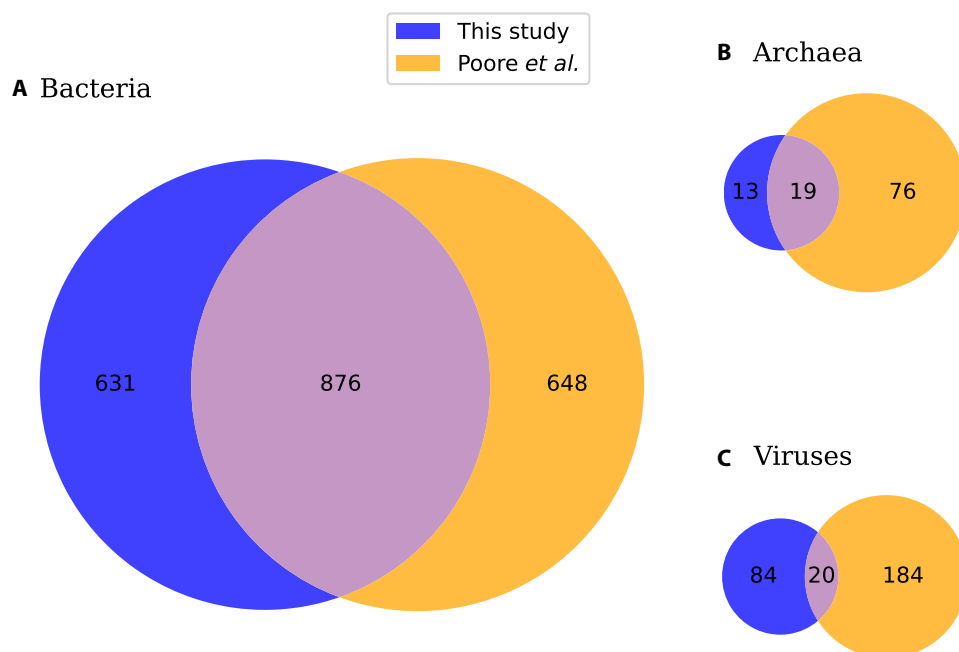
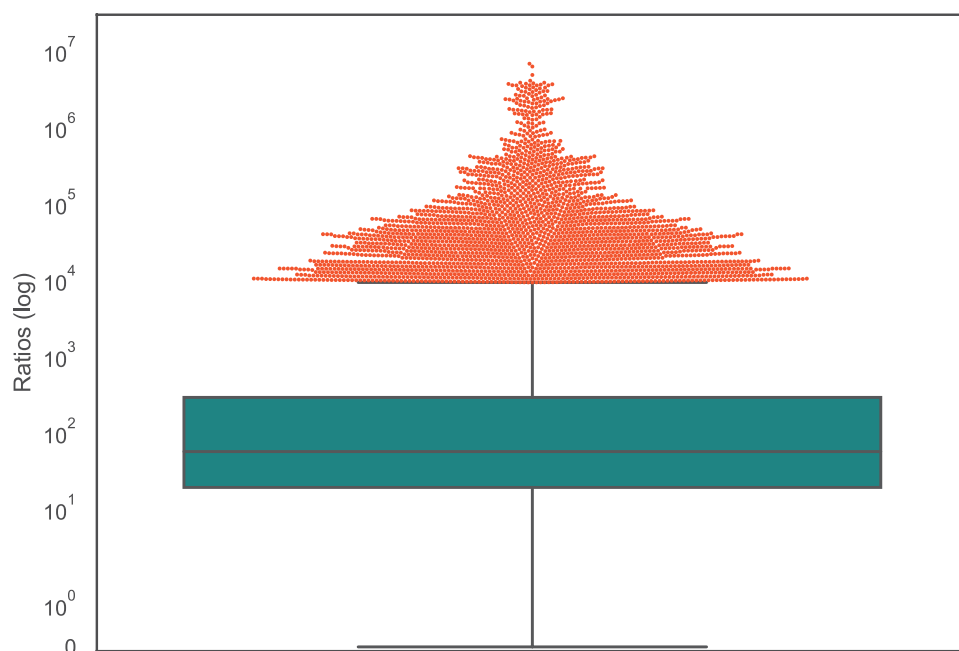


Fig. 3. Distribution of ratios of read counts between studies. The ratios between the numbers of microbial reads found by Poore *et al.* (1) and this study, for each sample and each microbial genus in which either study found at least 10 reads, are compared. The boxplot shows the median (56) and interquartile range of these ratios. The top 5% of points are shown as a cloud above the value 9186. Each point in the cloud represents the ratio, for a single sample and a single microbial genus, between the number of reads found by the two studies. The top 5% comprise 109,887 values, so for visualization purposes, only a random subset of 2500 points is shown.



most abundant species found exclusively in our analysis was *Schaalia turicensis* [previously called *Actinomyces turicensis* (12)], a bacterium that is commonly found in the oral and gut microbiome.

Comparison of fungal content to previous reports

In another recent study using the TCGA data, Narunsky-Haziza *et al.* (4) reported a strong association between mixtures of fungal species, which the authors called a “mycobiome,” and multiple cancer types. Of the 5734 WGS samples analyzed here, we identified 4271 that were identical to those used in the Narunsky-Haziza *et al.* study. We reanalyzed these samples for fungal content using a separate fungal genome database that contained 557 species, including all 224 of the species included in the previous study (4). The full set of read counts for the 4271 samples used in both studies can be found in data files S10 and S11, where the counts in data file S11 were taken from Narunsky-Haziza *et al.* (4). Because we used a superset of the fungal species, we identified reads from many species not reported in the previous study; however, these were observed at very low numbers, with an average read count of 3.8 reads per sample in species unique to our analysis.

Although our average read counts were in rough agreement, we found highly divergent results for a small number of species that were estimated by Narunsky-Haziza *et al.* to be highly abundant. These are illustrated in Fig. 5, which compares the maximum read counts from any sample for the top 10 most-abundant fungal species from the previous study (4). These include samples containing 2,013,180 reads from *Ramularia collo-cygni*, 656,503 reads from *Trichosporon asahii*, 101,344 reads from *Candida albicans*, and 54,641 reads from *Malassezia globosa*. In contrast, our counts for the same samples were 332, 4622, 266, and 4, values that range from 142-fold to 13,660-fold smaller.

Notably, the species with these large read counts were considered particularly important by the previous study, where they were used to define three fungi-driven “mycotypes” labeled F1 (*Malassezia*, *Ramularia*, and *Trichosporon*), F2 (*Candida* and *Aspergillus*), and F3 (multiple genera, including *Yarrowia*) that were reported to be associated with distinct immune responses and overall survival (4). Below we explain how contamination in the genomes themselves led to at least some of the higher read counts. We hypothesize that if the raw counts are corrected, the links between these mycotypes and cancer may be substantially weakened.

Contamination in the *M. globosa* genome

The highest read count for *M. globosa* as reported by Narunsky-Haziza *et al.* (4) was 54,641 reads from sample h2540, a blood-derived normal sample from the HNSC dataset, in which our reanalysis found only four *M. globosa* reads. We subsequently aligned all reads (without filtering) from sample h2540 against the *M. globosa* reference genome and found a very large number of matches, nearly all aligning to just two locations: an 897-bp contig and a 557-bp contig (Table 2). We then used BLAST (13, 14) to confirm that both contigs were human sequences that were mislabeled as *M. globosa*. The *M. globosa* genome assembly was revised in December 2023 (GenBank accession GCF_000181695.2), and the contigs shown in Table 2 were removed by National Center for Biotechnology Information (NCBI) because they were determined to be contaminated.

Contamination in the *T. asahii* genome

T. asahii also had an unusually high read count in the Narunsky-Haziza *et al.* study (4), which reported a maximum count of 656,503 reads in sample h1948 [a solid tissue normal sample from TCGA–lung

Fig. 4. Comparison of read counts for the top 10 most abundant microbial genera found in a previous study. Shown are the top 10 most-abundant genera ranked by the average number of reads reported in Poore *et al.* (1) across 4550 WGS samples from 25 cancer types. The left axis shows read counts as reported by Poore *et al.*, and the right axis shows read counts as computed in our reanalysis of the same samples. Note that the y axis scales differ by a factor of ~700. The x axis shows genus names.

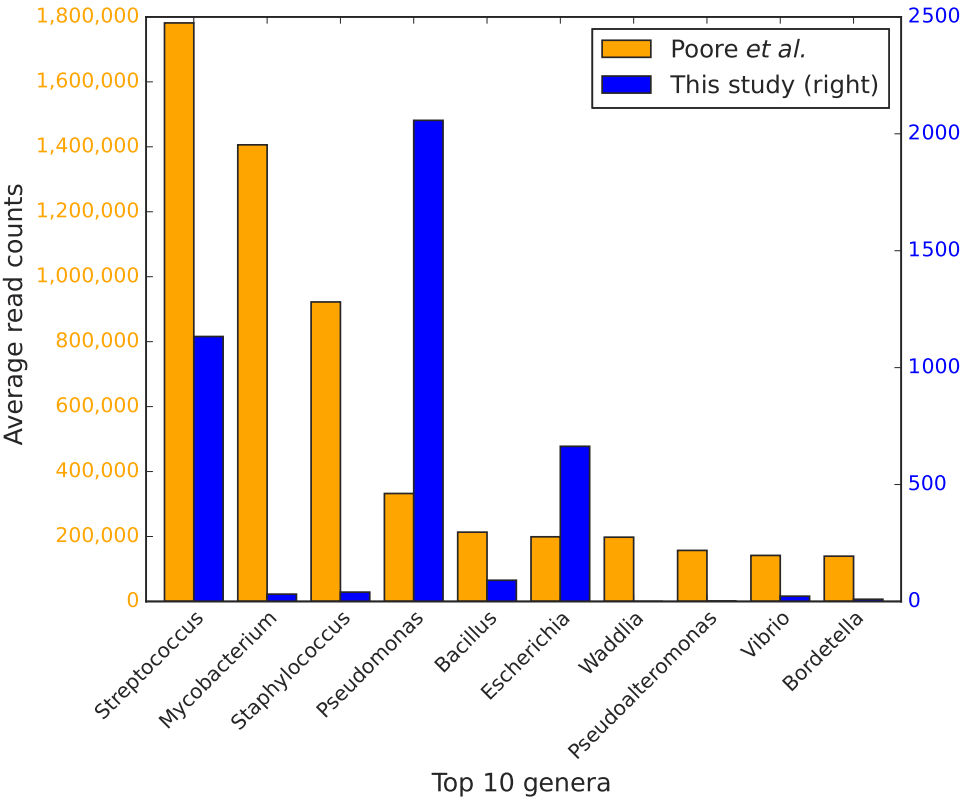


Fig. 5. Comparison of read counts in fungal species. Shown are the top 10 most-abundant fungal species ranked by the maximum number of reads reported in Narunsky-Haziza *et al.* (4), across 4271 WGS samples from 25 cancer types. The left axis shows read counts as reported by Narunsky-Haziza *et al.*, and the right axis shows read counts as computed in our reanalysis for the same samples and the same species. Note that the y axis scales differ by a factor of 50.

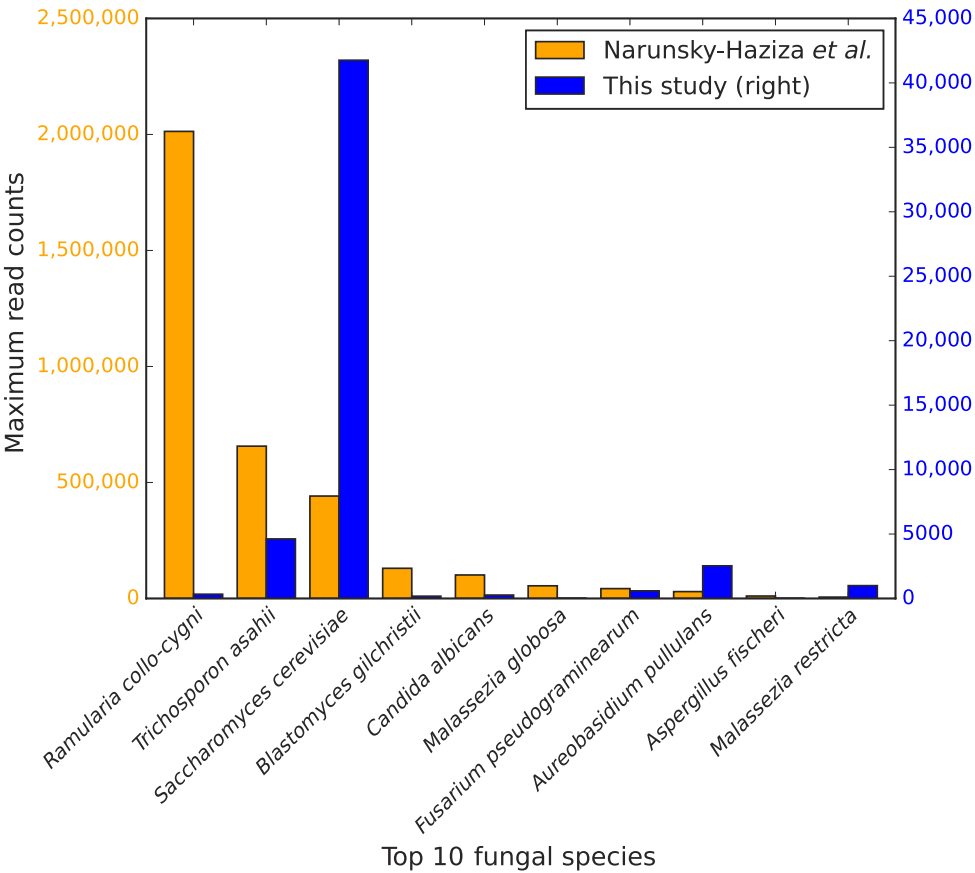


Table 2. Contaminants found in fungal genomes in GenBank. Columns 1 to 3 show the species name, GenBank accession number of the genome, and accession number of the scaffold in which the contaminant was found. Columns 4 to 6 show the starting location of the contaminant within the scaffold, its length, and the correct source of the DNA sequence.

Organism	Accession	Scaffold accession	Start	Length	Contamination source
<i>M. globosa</i>	GCF_000181695.1	NW_001849834.1	1	897	Human
<i>M. globosa</i>	GCF_000181695.1	NW_001849877.1	1	557	Human
<i>T. asahii</i>	GCF_000293215.1	NW_014040855.1	924,646	58	Vector: NGB00360.1, Illumina PCR primer
<i>T. asahii</i>	GCF_000293215.1	NW_014040868.1	203,990	64	Vector: NGB00852.1, NEBNext Index 6 primer
<i>R. collo-cygni</i>	GCF_900074925.1	NW_019716264.1	394,971	58	Vector: NGB00360.1, Illumina PCR primer
<i>R. collo-cygni</i>	GCF_900074925.1	NW_019716256.1	1,478,660	58	Vector: NGB00360.1, Illumina PCR primer

adenocarcinoma (LUAD)]. Upon aligning all reads (unfiltered) from that sample to the *Trichosporon* genome, we found an even higher number of matches, over 54 million; however, 99.99% of the matches hit the same 80-bp interval in the genome. We investigated and found that 58 bp from this 80-bp sequence was identical to an Illumina sequencing vector (accession NGB00360.1); in other words, it is a contaminant in the genome assembly. The vector contaminant occurs in the middle of a large, 2.7-Mbp scaffold (NCBI accession NW_014040855.1, see Table 2). In addition, sample h1948 was a failed sequencing run, in which 95.4% of the read pairs (160 million of a total of 179 million) were vector derived.

We found similar results for other samples; for example, Narunsky-Haziza *et al.* (4) reported 302,882 *Trichosporon* reads in sample h1325, a lung cancer tumor sample. When we aligned the entire set of reads (unfiltered) from this sample to *T. asahii*, we found 12,235,713 matching reads, and all except for 18 reads aligned to the same 58-bp vector contaminant mentioned above. In a different sample, we found 1980 reads matching a 64-bp interval in the *T. asahii* genome that turned out to be another vector (accession NGB00852.1), also shown in Table 2. Thus, for these and other samples, the large numbers of apparent matches to *Trichosporon* appear to have been the result of vector contaminants in the fungal genome sequence.

Contamination in the *Ramularia collo-cygni* genome

Sample h1948, a normal tissue sample from the lung cancer (LUAD) dataset, was reported to contain 2,013,180 reads from *R. collo-cygni* (4), whereas we found only 322 reads. We realigned the filtered reads to the *R. collo-cygni* genome using the same Bowtie2 parameters as used by Narunsky-Haziza *et al.* and found ~34 million matching reads, all matching just two locations: a 77-bp sequence on RCC_scaffold10 (NCBI accession NW_019716264.1) and a 77-bp sequence on RCC_scaffold02 (NCBI accession NW_019716256.1, see Table 2). Both of these sequences contain a 58-bp subsequence identical to an Illumina PCR primer, one of the two that we found in *Trichosporon*. Thus, these apparent matches to *Ramularia* appear to have been the result of vector contamination.

Analysis of high read counts for *C. albicans*

The highest read count reported by Narunsky-Haziza *et al.* (4) for *C. albicans* was 101,344 reads from sample h949, a primary tumor sample from the HNSC dataset. We attempted to replicate this finding by processing the sample using the host depletion pipeline described in the original study, which left only 93,556 reads, a number already lower than the reported count for *C. albicans*. We aligned these reads to the *C. albicans* genome using Bowtie2 (15), which detected only 230 matching reads. Using the same procedures, we analyzed the sample with the second highest *C. albicans* read count, h5103, a normal tissue sample from the lung cancer [lung squamous cell carcinoma (LUSC)] dataset with 75,799 reported matches, and found only 544 matching reads. We were similarly unable to replicate high read counts for *C. albicans* in other samples.

To attempt to explain the far higher read counts reported by Narunsky-Haziza *et al.* (4), we aligned the entire set of unfiltered reads (that is, without first removing human-matching reads) from the HNSC and LUSC samples to the *C. albicans* genome, which yielded 75,392 and 133,576 matching reads, respectively, values closer to the original report. However, 91% of the 75,392 reads from the HNSC sample were originally mapped to human in the raw TCGA data and thus should not have been included in the “nonhuman” read sets. We investigated further and determined that nearly all these reads matched ribosomal RNA (rRNA) genes in both human and *C. albicans*, although the matches to *C. albicans* had much higher similarity. This analysis suggests that *C. albicans* was genuinely present in the samples (whether it was in the tissues or a contaminant), but it is unclear how the pipeline in the original study yielded these high read counts.

Last, we note that Narunsky-Haziza *et al.* (4) reported a maximum of ~441,000 reads from one sample matching *S. cerevisiae*, a finding that we were unable to replicate (Fig. 5). In our analysis of the same sample, we found ~40,000 reads matching *S. cerevisiae*. These matches predominantly fell within a region of the yeast genome that contains the 18S rRNA gene and that has substantial similarity to human, suggesting that these were human reads that slipped past the filtering step.

DISCUSSION

Using updated computational methods and databases, we have analyzed the nonhuman content in a large collection of WGS datasets in TCGA covering 25 distinct types of cancer and created a comprehensive dataset encompassing detailed read counts for bacteria, viruses, archaea, fungi, and other microbes in 5734 samples from

tumors and normal tissue. Our data show that most of the read counts reported in a previously published dataset are greatly inflated, often exceeding the counts we obtained by factors exceeding 1000. As we explained previously (2), these overcounts can be attributed in part to the inclusion of numerous draft bacterial genomes (which themselves contain contaminants) in the database used for the metagenomic analysis. Although the primary paper was recently retracted, the results from Poore *et al.* (1) have been used in at least a dozen other studies (16–27) that downloaded their read count data and then published findings based on that data, and another recent study (28) similarly based its results on the “mycobiome” data from Narunsky-Haziza *et al.* (4). Our analyses here used a cleaner genome database, which yielded much lower read counts for the microbial species we identified. We hope that by providing a more accurate set of read counts for the same TCGA samples, our dataset can serve as a valuable scientific resource, enabling researchers to better distinguish genuine microbial signals from background noise and contaminants in future investigations.

In this context, it is worth noting that all tumor types are not equal and that the biology and anatomical location of a tumor have a large impact on a tumor’s accessibility to microbes. For example, tumors in the oral cavity and colon will be exposed to far more bacteria than tumors from internal organs such as the brain. This factor should be considered when trying to determine whether certain microbes are contaminants.

We also extended our previous work to add comparisons to the fungal genomes found in cancer by Narunsky-Haziza *et al.* (4). Upon looking at the data published in that study, we found that several of the key fungal species used to create cancer-specific signatures had excessively high read counts and that at least some of those counts were the result of contamination in the fungal genome sequences. The contaminants included both human DNA and vector contaminants, either of which can lead to high numbers of false positives when doing metagenomic analysis.

Our study has limitations. One is that we did not further examine the numerous instances where we found microbes in the TCGA samples that might be more than just contaminants. Among those microbes, a small number of abundant species might represent acute infections from the original patient samples, although we were not able to test that hypothesis. For example, *Metamycoplasma salivarium*, an oral pathogen that may be pathogenic in immunocompromised individuals, was the top species in esophageal carcinoma (29), and this finding might merit follow-up investigation using new data. Another limitation is that we did not investigate the specific question of whether a microbiome, that is, a community of microbes, was present in any of the cancer types. However, the datasets provided here contain numerous species and ensembles of species, in a variety of cancer types, that might be worthy of further study.

Multiple other studies have recently reported microbiomes in cancer, including a widely publicized report of a fungal mycobiome in pancreatic cancer (30) that was only recently (in 2023) shown to be flawed (31), likely because of misidentification of fungal reads in the original data. Similarly, a 2020 study reported intracellular microbes in seven types of cancer (32), but an attempt to replicate those findings in breast cancer found no evidence of microbes in cancer cells, only inside ducts or macrophages (33). Most recently, a study involving 22 cancer types found evidence for distinctive microbial communities only in colorectal tumors but not in other tumor types (34). Together with the results described here, these

reports suggest that claims regarding microbiomes and cancer need to be scrutinized more rigorously than they have been in the recent past and that contamination of human samples with environmental microbes can easily be mistaken for a genuine signal.

Last, we did find a number of individual microbes present in high abundance in some samples. Some of these were consistent with prior knowledge, whereas others may be worthy of further investigation, ideally with additional, independent experimental data.

MATERIALS AND METHODS

Study design

Our research objectives were to determine the microbial content present in a large collection of human tissue samples sequenced as part of TCGA project; to compare our findings with previously published reports based on the same samples; and to release detailed data describing the viruses, bacteria, and fungi identified in those samples. A further goal was to use rigorous computational methods and carefully curated reference databases to minimize errors and thereby to provide a data resource that future studies could use to explore hypotheses related to microbes or microbiomes that might be associated with cancer. We began by collecting raw data from all TCGA samples (as of mid-2024) for which TCGA had generated WGS data, which comprised 25 cancer types and 5734 samples including tumors, matching normal tissue, and circulating blood.

Data and analysis

We downloaded sequence data from the Genome Data Commons at the US National Cancer Institute (gdc.cancer.gov) for 25 cancer types from the TCGA project. Metadata on all samples used in this study and previous studies, including unique sample identifiers from TCGA, can be found in data file S12. In total, data from 5734 WGS samples were downloaded from the National Cancer Institute data portal, which had aligned the reads to either hg19 or GRCh38 using bwa (8). Of these samples, 2824 represented solid tumors, 569 were solid normal tissue, 64 were blood-derived cancer, and 2277 were blood-derived normal. We downloaded all reads that were “unmapped” in the TCGA data, where their reference genomes included the human genome (either hg19 or GRCh38, depending on the date of data collection) as well as human papilloma virus (multiple subtypes), hepatitis C virus 1 and 2, hepatitis B, and Epstein-Barr virus. We separately downloaded all reads identified as viral by TCGA and added them to the unmapped reads for each sample. We then aligned the unmapped reads against the CHM13 human reference using Bowtie2 (15) to remove more human reads (Table 1).

We classified these two-pass-filtered read sets using KrakenUniq (7) using its default parameters in paired-read mode, which treats each pair of reads as a single discontinuous sequence. KrakenUniq is a k-mer-based metagenomic classifier that has repeatedly been shown (35) to be among the most accurate methods available for this task. We classified the ~15 billion two-pass-filtered read pairs (Table 1) against a curated database, Microbial2023, containing all RefSeq bacterial, archaeal, and viral complete genomes; a collection of curated eukaryotic pathogen genomes from EuPathDB54; 10,798 vector sequences from UniVec and EmVec; the CHM13v2.0 genome; and the GRCh38.p14 human genome. In total, Microbial2023 contains 50,079 genomes (29,798 species) divided into 34,452 bacteria (16,150 species), 14,018 viruses (12,910 species), 534 archaea (422 species), and 503 eukaryotic microbes (316 species). The list of

species and their GenBank accessions can be found in data file S13. The Microbial2023 database, which is 535 GB in size, can be downloaded from <https://benlangmead.github.io/aws-indexes/k2>. A total of 891,242,697 nonvector, nonhuman reads were classified by KrakenUniq at the genus level or below; of these, 827,851,151 were classified at the species level or below. Note that ~63.4 million reads were classified at the genus level but not at the species level because they matched more than one species equally well.

To conduct a more comprehensive search against fungi, we created a separate database containing all 572 fungal genomes (557 species) in the NCBI RefSeq database as of mid-2023, which we designate as Fungi_RefSeq. The list of species and their GenBank accessions can be found in data file S14. After processing each sample using Microbial2023, we extracted all reads that either failed to match or were classified as fungi (taxid = 4751) and screened them against Fungi_RefSeq using KrakenUniq with default parameters. These steps yielded read counts for all 5734 samples against the 557 fungal species.

To compare our bacterial, archaeal, and viral read counts to Poore *et al.*'s results, we compared the sample metadata and species and determined that 4550 of 5734 TCGA samples were analyzed in both studies and 1289 of 1993 microbial genera were reported in both studies. Note that because Microbial2023 is a newer database, it has some species missing from the Poore *et al.* study. Conversely, because Microbial2023 only includes finished genomes, whereas Poore *et al.* included draft genomes, some species and genera used in Poore *et al.* are missing from Microbial2023. Similarly for Narunsky-Haziza *et al.*'s results, we matched 4271 of 5734 TCGA samples. All 224 fungal species used in Narunsky-Haziza *et al.* were included in our reanalysis. A list of species name changes from RefSeq200 to RefSeq220 involving some of these fungi can be found in data file S15.

In our analysis of the high read counts in *M. globosa*, *T. asahii*, *R. collo-cygni*, and *C. albicans*, we aligned human-filtered reads from the samples against their respective reference genomes using the same Bowtie2 parameters as used by Narunsky-Haziza *et al.* (--end-to-end --very-sensitive -k 16 --np 1 --mp 1,1 --rdg 0,1 --rfg 0,1 --score-min L,0,-0.05).

Statistical analysis

Numeric data are described in the figures and results text.

Supplementary Materials

The PDF file includes:

Figs. S1 and S2

Other Supplementary Material for this manuscript includes the following:

Data files S1 to S15

MDAR Reproducibility Checklist

REFERENCES AND NOTES

- G. D. Poore, E. Kopylova, Q. Zhu, C. Carpenter, S. Fraraccio, S. Wandro, T. Kosciolk, S. Janssen, J. Metcalf, S. J. Song, J. Kanbar, S. Miller-Montgomery, R. Heaton, R. McKay, S. P. Patel, A. D. Swafford, R. Knight, Microbiome analyses of blood and tissues suggest cancer diagnostic approach. *Nature* **579**, 567–574 (2020).
- A. Gihawi, Y. Ge, J. Lu, D. Pui, A. Xu, C. S. Cooper, D. S. Brewer, M. Perlea, S. L. Salzberg, Major data analysis errors invalidate cancer microbiome findings. *mBio* **14**, e0160723 (2023).
- G. D. Poore, E. Kopylova, Q. Zhu, C. Carpenter, S. Fraraccio, S. Wandro, T. Kosciolk, S. Janssen, J. Metcalf, S. J. Song, J. Kanbar, S. Miller-Montgomery, R. Heaton, R. McKay, S. P. Patel, A. D. Swafford, R. Knight, Retraction Note: Microbiome analyses of blood and tissues suggest cancer diagnostic approach. *Nature* **631**, 694 (2024).

4. L. Narunsky-Haziza, G. D. Sepich-Poore, I. Livyatan, O. Asraf, C. Martino, D. Nejman, N. Gavert, J. E. Stajich, G. Amit, A. González, S. Wandro, G. Perry, R. Ariel, A. Meltser, J. P. Shaffer, Q. Zhu, N. Balint-Lahat, I. Barshack, M. Dadiani, E. N. Gal-Yam, S. P. Patel, A. Bashan, A. D. Swafford, Y. Pillel, R. Knight, R. Straussman, Pan-cancer analyses reveal cancer-type-specific fungal ecologies and bacteriome interactions. *Cell* **185**, 3789–3806. e17 (2022).
5. F. P. Breitwieser, M. Pertea, A. V. Zimin, S. L. Salzberg, Human contamination in bacterial genomes has created thousands of spurious proteins. *Genome Res.* **29**, 954–960 (2019).
6. M. Dreyfus, P. Régnier, The poly(A) tail of mRNAs: Bodyguard in eukaryotes, scavenger in bacteria. *Cell* **111**, 611–613 (2002).
7. F. P. Breitwieser, D. N. Baker, S. L. Salzberg, KrakenUniq: Confident and fast metagenomics classification using unique k-mer counts. *Genome Biol.* **19**, 198 (2018).
8. H. Li, R. Durbin, Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* **25**, 1754–1760 (2009).
9. The Cancer Genome Atlas Research Network, Integrated genomic and molecular characterization of cervical cancer. *Nature* **543**, 378–384 (2017).
10. A. Sasaki, M. Miyaniishi, K. Ozaki, M. Onoue, K. Yoshida, Molecular characterization of a partitivirus from the plant pathogenic ascomycete *Rosellinia necatrix*. *Arch. Virol.* **150**, 1069–1083 (2005).
11. S. Chiba, Y.-H. Lin, H. Kondo, S. Kanematsu, N. Suzuki, A novel victorivirus from a phytopathogenic fungus, *Rosellinia necatrix*, is infectious as particles and targeted by RNA silencing. *J. Virol.* **87**, 6727–6738 (2013).
12. I. Nouioui, L. Carro, M. García-López, J. P. Meier-Kolthoff, T. Woyke, N. C. Kyrpides, R. Pukall, H.-P. Klenk, M. Goodfellow, M. Göker, Genome-based taxonomic classification of the Phylum *Actinobacteria*. *Front. Microbiol.* **9**, 2007 (2018).
13. Z. Zhang, S. Schwartz, L. Wagner, W. Miller, A greedy algorithm for aligning DNA sequences. *J. Comput. Biol.* **7**, 203–214 (2000).
14. A. Morgulis, G. Coulouris, Y. Raytselis, T. L. Madden, R. Agarwala, A. A. Schäffer, Database indexing for production MegaBLAST searches. *Bioinformatics* **24**, 1757–1764 (2008).
15. B. Langmead, S. L. Salzberg, Fast gapped-read alignment with Bowtie 2. *Nat. Methods* **9**, 357–359 (2012).
16. L. C. Hermida, E. M. Gertz, E. Rupp, Predicting cancer prognosis and drug response from the tumor microbiome. *Nat. Commun.* **13**, 2896 (2022).
17. S. Parida, S. Siddharth, Y. Xia, D. Sharma, Concomitant analyses of intratumoral microbiota and genomic features reveal distinct racial differences in breast cancer. *NPJ Breast Cancer* **9**, 4 (2023).
18. A. W. Mao, H. Barck, J. Young, A. Paley, J.-H. Mao, H. Chang, Identification of a novel cancer microbiome signature for predicting prognosis of human breast cancer patients. *Clin. Transl. Oncol.* **24**, 597–604 (2022).
19. M. Luo, Y. Liu, L. C. Hermida, E. M. Gertz, Z. Zhang, Q. Li, L. Diao, E. Rupp, L. Han, Race is a key determinant of the human intratumor microbiome. *Cancer Cell* **40**, 901–902 (2022).
20. G. Zhu, H. Su, C. H. Johnson, S. A. Khan, H. Kluger, L. Lu, Intratumour microbiome associated with the infiltration of cytotoxic CD8+ T cells and patient survival in cutaneous melanoma. *Eur. J. Cancer* **151**, 25–34 (2021).
21. F. Chen, J. Yang, Y. Guo, D. Su, Y. Sheng, Y. Wu, Integrating bulk and single-cell RNA sequencing data reveals the relationship between intratumor microbiome signature and host metabolic heterogeneity in breast cancer. *Front. Immunol.* **14**, 1140995 (2023).
22. C. Chen, Y. Cai, Y. Liu, S. Chen, Y. Li, F. Zhang, M. Zhang, Z. Lu, P. Ying, J. Huang, L. Fan, X. Cai, C. Ning, W. Wang, Y. Jiang, H. Zhang, S. Yang, Z. Wang, X. Wang, S. Zhang, C. Huang, B. Xu, Z. Fu, Q. Song, M. Jin, K. Chen, H. Chen, M. Dai, X. Miao, X. Yang, Y. Zhu, J. Tian, Pan-cancer analysis of microbiome quantitative trait loci. *Cancer Res.* **82**, 3449–3456 (2022).
23. D. M. Lim, H. Lee, K. Eom, Y. H. Kim, S. Kim, Bioinformatic analysis of the obesity paradox and possible associated factors in colorectal cancer using TCGA cohorts. *J. Cancer* **14**, 322–335 (2023).
24. R. Bentham, K. Litchfield, T. B. K. Watkins, E. L. Lim, R. Rosenthal, C. Martínez-Ruiz, C. T. Hiley, M. A. Bakir, R. Salgado, D. A. Moore, M. Jamal-Hanjani, TRACERx Consortium, C. Swanton, N. McGranahan, Using DNA sequencing data to quantify T cell clonal and therapy response. *Nature* **597**, 555–560 (2021).
25. Y. K. Kim, E. J. Kwon, Y. Yu, J. Kim, S.-Y. Woo, H.-S. Choi, M. Kwon, K. Jung, H.-S. Kim, H. R. Park, D. Lee, Y. H. Kim, Microbial and molecular differences according to the location of head and neck cancers. *Cancer Cell Int.* **22**, 135 (2022).
26. Y. Xu, J. Zhao, Y. Ma, J. Liu, Y. Cui, Y. Yuan, C. Xiang, D. Ma, H. Liu, The microbiome types of colorectal tissue are potentially associated with the prognosis of patients with colorectal cancer. *Front. Microbiol.* **14**, 1100873 (2023).
27. Y. Li, D. Zhang, M. Wang, H. Jiang, C. Feng, Y.-X. Li, Intratumoral microbiota is associated with prognosis in patients with adrenocortical carcinoma. *iMeta* **2**, e102 (2023).
28. S.-W. Guan, Q. Lin, X.-D. Wu, H.-B. Yu, Weighted gene coexpression network analysis and machine learning reveal oncogene associated microbiome plays an important role in tumor immunity and prognosis in pan-cancer. *J. Transl. Med.* **21**, 537 (2023).
29. E. Yacoub, O. M. Saed Abdul-Wahab, M. H. Al-Shyarba, B. Ben Abdelmoumen Mardassi, The relationship between *Mycoplasmas* and cancer: Is it fact or fiction? Narrative review and update on the situation. *J. Oncol.* **2021**, 1–13 (2021).
30. B. Aykut, S. Pushalkar, R. Chen, Q. Li, R. Abengozar, J. I. Kim, S. A. Shadalooy, D. Wu, P. Preiss, N. Verma, Y. Guo, A. Saxena, M. Vardhan, B. Diskin, W. Wang, J. Leinwand, E. Kurz, J. A. Kochen Rossi, M. Hundeyin, C. Zambirinis, X. Li, D. Saxena, G. Miller, The fungal mycobiome promotes pancreatic oncogenesis via activation of MBL. *Nature* **574**, 264–267 (2019).
31. A. A. Fletcher, M. S. Kelly, A. M. Eckhoff, P. J. Allen, Revisiting the intrinsic mycobiome in pancreatic cancer. *Nature* **620**, E1–E6 (2023).
32. D. Nejman, I. Livyatan, G. Fuks, N. Gavert, Y. Zwing, L. T. Geller, A. Rotter-Maskowitz, R. Weiser, G. Malle, E. Gigi, A. Meltser, G. M. Douglas, I. Kamer, V. Gopalakrishnan, T. Dadosh, S. Levin-Zaidman, S. Avnet, T. Atlan, Z. A. Cooper, R. Arora, A. P. Cogdill, M. A. W. Khan, G. Ologun, Y. Bussi, A. Weinberger, M. Lotan-Pompan, O. Golani, G. Perry, M. Rokah, K. Bahar-Shany, E. A. Rozeman, C. U. Blank, A. Ronai, R. Shaoul, A. Amit, T. Dorfman, R. Kremer, Z. R. Cohen, S. Harnof, T. Siegal, E. Yehuda-Shnaidman, E. N. Gal-Yam, H. Shapira, N. Baldini, M. G. I. Langille, A. Ben-Nun, B. Kaufman, A. Nissan, T. Golan, M. Dadiani, K. Levanon, J. Bar, S. Yust-Katz, I. Barshack, D. S. Peeper, D. J. Raz, E. Segal, J. A. Wargo, J. Sandbank, N. Shental, R. Straussman, The human tumor microbiome is composed of tumor type-specific intracellular bacteria. *Science* **368**, 973–980 (2020).
33. N. F. C. de Miranda, V. T. H. B. M. Smit, M. van der Ploeg, J. Wesseling, J. Neefjes, Absence of lipopolysaccharide (LPS) expression in breast cancer cells. *Cell. Oncol.* **48**, 1123–1126 (2025).
34. A. Gihawi, H. M. Wood, J. Clark, J. O'Grady, R. A. Eeles, D. C. Wedge, G. M. Jakobsdottir, G. Magiorkinis, A. G. Schache, L. Masterson, M. Lechner, T. R. Fenton, T. M. Jones, A. Flanagan, S. De Noon, A. Rubinstein, R. Hurst, C. S. Cooper, D. S. Brewer, The landscape of microbial associations in human cancer. *Sci. Transl. Med.* **17**, eads6166 (2025).
35. J. Marić, K. Krizanović, S. Riondet, N. Nagarajan, M. Šikić, Comparative analysis of metagenomic classifiers for long-read sequencing datasets. *BMC Bioinformatics* **25**, 15 (2024).

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Comprehensive analysis of microbial content in whole-genome sequencing samples from The Cancer Genome Atlas project

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Editor's summary

Whether microbes are associated with cancer, and if so which cancers, has been a controversial matter. Ge *et al.* reanalyzed genome sequencing data from The Cancer Genome Atlas, a commonly used data source for analyzing both cancer cell and bacterial, viral, and fungal sequences isolated from patients. After rigorously eliminating microbial sequences that were likely only contaminants and not true signals, they found evidence for limited associations of the microbiome with cervical and gastrointestinal cancers. The authors' processed TCGA data are publicly available and provide a useful resource for researchers studying the cancer microbiome. —Catherine Charneski

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